Higher-Degree Polynomial Families on $x^4 + y^3 + z^2 + 1 = 0$: Quartic Divisorial Syzygies for $\deg x = 4$, Resolution of the $\nu \in \{4, 5\}$ Systems, Modulo-8 Artin–Schreier & Modulo-9 Frobenius Obstructions in $\mathbb{Z}[t]$, and General Bounds

Research Continuation Report

Extending *Integer solutions of* $x^4 + y^3 + z^2 + 1 = 0$ (15–17 October 2026)

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Abstract

We extend the algebraic, geometric, and arithmetic theory of polynomial solutions $(x(t), y(t), z(t))$ to the affine surface $S : x^4 + y^3 + z^2 + 1 = 0$ (equivalently $x^4 - Y^3 + z^2 + 1 = 0$ or $Y^3 - z^2 = x^4 + 1$ with $Y = -y$) across both $\mathbb{Q}[t]^3$ and $\mathbb{Z}[t]^3$ (as well as $\mathbb{Z}_+[t]^3$) for arbitrary degree $\deg x \geq 1$. Building on the rational elliptic fibration $\pi : \hat{E} \to \mathbb{P}^1_x$ of Mordell–Weil lattice E_6^* with $\mathrm{MW}(\hat{E}/\mathbb{Q}(x)) = \{O\}$, we establish twelve principal results: (1) universal positivity lower bounds $\deg Y \geq \lceil 4d_x/3 \rceil$ and canonical Riemann–Roch trace-zero divisorial reductions for all $d_x = \deg x \geq 1$; (2) universal Davenport–Stothers–Zannier and Mason–Stothers upper bounds confining all nonconstant solutions for $d_x \geq 3$ to either the canonical regime $(3k, 4k, 6k)$ or the Hall–Davenport regime $(d_x, 2\nu, 3\nu)$ with $\deg Y \leq 8d_x - 2$ (unconditional over $\mathbb{Q}$ for $d_x \in \{3, 4\}$) and $\deg Y \leq 10d_x - 4$ unconditionally for all $d_x \geq 3$; (3) a universal *Mutual Rational Root Locking Theorem* over $\mathbb{Q}[t]$ proving via the rank-0 Mordell curve $E_0(\mathbb{Q}) = \{O, (1, 0)\}$ ($z^2 = Y^3 - 1$) that $x(r) = 0 \iff z(r) = 0 \implies Y(r) = 1$ for every $r \in \mathbb{Q}$, with even vanishing multiplicity $\mathrm{ord}_{t=r}(Y(t) - 1) = \min(4\,\mathrm{ord}_{t=r}\, x, 2\,\mathrm{ord}_{t=r}\, z) \geq 2$; (4) an explicit extension of the trace-zero divisorial method to $\deg x = 4$, proving by UFD coprimality that $B(x)z = Y^2 + m(x)Y + n(x)$ has constant Y^2-coefficient and reduces to the coupled polynomial syzygy system $m^3 - e_2 m - e_3 \equiv 0$ and $(e_2 - m^2)^2 + 4(e_1 + 2m)(x^4 + 1) - 4e_4 \equiv 0$ with $\deg_x B \leq \lfloor \nu/4 \rfloor$, $\deg_x m \leq \lfloor \nu/2 \rfloor$, and $\deg_x n = \nu$; (5) the binary cubic covariant syzygy $4p^3 + 27N^2 = (27u^2 + 4s^3)I^2$ and $\mathbb{G}_m^2$-bihomogeneous reduction for $\deg x = 3$; (6) complete resolution of the open $\nu = 4$ ($\deg x = 3, \deg Y = 8, \deg z = 12$) system over $\mathbb{Q}$ and discovery of an explicit $(3, 8, 12)$ polynomial identity over $\mathbb{Q}(\sqrt{3}, 2^{1/3})$; (7) complete reduction and exclusion of the $\nu = 5$ ($\deg x = 3, \deg Y = 10, \deg z = 15$) system across both $m_1 = 0$ and $m_1 \neq 0$ branches via denominator-free coordinates $b_2 = (s - 1)B_2$, two-step Euclidean elimination $L = -A_0(\tilde{B}_1, s)/A_1(\tilde{B}_1, s)$, and the overdetermined curve $H_9(\tilde{B}_1, s) = 0$; (8) infinite towers of polynomial identities over $\mathbb{Q}(\sqrt{3}, 2^{1/3})$ of degrees $(3k, 4k, 6k)$ and $(3k, 8k, 12k)$ from the D_4^* pullback elliptic surface $\mathcal{E} : \tilde{z}^2 = \tilde{Y}^3 - (w^3 + 1)$; (9) strict equivalence between polynomial solutions in $\mathbb{Z}[t]^3$ and $\mathbb{Z}_+[t]^3$; (10) the Modulo-4 Frobenius and $\mathbb{F}_2[t]$-Derivative Obstruction $8 \mid \deg_{\mathbb{F}_2}(Y \bmod 2) \leq 2\deg x - 2$ (for $\bar{w} \neq 1$) and $Y(t) \equiv 1 + 2x(t)^4 \pmod 4$ (for $\bar{w} = 1$, unconditional for $\deg x \leq 4$); (11) a new *Modulo-8 Artin–Schreier and Modulo-16 2-Adic Degree Jump Theorem* in $\mathbb{Z}[t]$ proving that $\bar{Y}_2 = \bar{x}^4 \bar{x}_{\mathrm{AS}} + \bar{W}^2 + \bar{x}^2 \bar{W}$ in $\mathbb{F}_2[t]$, which forces $\deg Y \geq 4m_x + m_{\mathrm{AS}} + 1$ (and $\deg Y \geq 5m_x + 1$ whenever $m_x = \deg_{\mathbb{F}_2}(x \bmod 2)$ is odd, immediately ruling out $\deg Y = 14$ for $\deg x = 3$ with $\mathrm{LC}(x)$ odd), together with a *Modulo-9 Frobenius Carry Obstruction* in $\mathbb{F}_3[t]$; and (12) general conditional nonexistence theorems for all $\deg x \geq 3$ under the Polynomial Hall–Vojta Conjecture and Severi stratum codimension bounds.

Result	Tag	Summary and Section Reference
Theorem 2.1 (Lower Bound & Trace)	[E]	For all $d_x = \deg x \geq 1$, $\deg Y \geq \lceil 4d_x/3 \rceil$ over $\mathbb{R}$, and $\sum_{i=1}^{d_x}[P_i] = O$ in $E(\mathbb{Q}(x))$ yields canonical divisorial reduction (Section 2.1).
Theorem 2.2 (Universal Upper Bound)	[E]	For all $d_x \geq 3$, $\deg Y = 2\nu \leq 8d_x - 2$ (when $\gcd(Y, z) = 1$, automatic for $d_x \in \{3, 4\}$) and $\deg Y \leq 10d_x - 4$ unconditionally (Section 2.2).
Theorem 2.3 (Rational Root Locking)	[E]	For all $d_x \geq 1$ and $r \in \mathbb{Q}$, $x(r) = 0 \Leftrightarrow z(r) = 0 \Rightarrow Y(r) = 1$ via $E_0(\mathbb{Q}) = \{O, (1, 0)\}$, with $\mathrm{ord}_{t=r}(Y - 1) = \min(4e_x, 2e_z) \geq 2$ even (Section 2.3).
Theorem 3.1 ($\deg x = 4$ Syzygies)	[E]	UFD coprimality forces $q(x) = 1$ in $B(x)z = Y^2 + m(x)Y + n(x)$, reducing $\deg x = 4$ to $m^3 - e_2 m - e_3 = 0$ and $(e_2 - m^2)^2 + 4(e_1 + 2m)(x^4 + 1) = 4e_4$ with $\deg_x B \leq \lfloor \nu/4 \rfloor$, $\deg_x m \leq \lfloor \nu/2 \rfloor$, $\deg_x n = \nu$ (Section 3).
Theorem 4.1 (Covariant Syzygy)	[E]	$p = \frac{1}{12}\mathrm{Hess}(I)$ and $N = -\frac{1}{9}\mathrm{Jac}(I, p)$ satisfy $4p^3 + 27N^2 = (27u^2 + 4s^3)I^2$; $\Phi(u) \equiv 0$ is $\mathbb{G}_m^2$-bihomogeneous in degrees $u^{\geq 5}$ (Section 4.1).
Theorems 4.2–4.3 ($\nu = 4$ Resolution)	[E]/[C]	Explicit $(3, 8, 12)$ curve over $\mathbb{Q}(\sqrt{3}, 2^{1/3})$ and complete exclusion of $\deg x = 3, \deg Y = 8, \deg z = 12$ over $\mathbb{Q}$ in both $m_1 = 0$ and $m_1 \neq 0$ branches (Section 4.2).
Theorems 4.4–4.5 ($\nu = 5$ Resolution)	[E]/[C]	Complete reduction and exclusion of $\deg x = 3, \deg Y = 10, \deg z = 15$ via $Q_5(\lambda) = 0$ ($m_1 = 0$) and denominator-free Euclidean reduction to $H_9(\tilde{B}_1, s) = 0$ ($m_1 \neq 0$) (Section 4.3).
Theorem 5.1 (Pullback Towers $d_x = 3k$)	[E]	Pullback to $\mathcal{E} : \tilde{z}^2 = \tilde{Y}^3 - (w^3 + 1)$ of lattice D_4^* gives infinite families of degrees $(3k, 4k, 6k)$ and $(3k, 8k, 12k)$ over $\mathbb{Q}(\sqrt{3}, 2^{1/3})$ for all $k \geq 1$ (Section 5).
Theorems 6.1–6.2 ($\mathbb{Z}_+[t]$ & Mod-4)	[E]	Equivalence $\mathbb{Z}[t]^3 \Leftrightarrow \mathbb{Z}_+[t]^3$; Modulo-4 Frobenius & derivative bound $8 \mid \deg_{\mathbb{F}_2}(Y \bmod 2) \leq 2d_x - 2$ and $Y \equiv 1 + 2x^4 \pmod 4$ for $\bar{w} = 1$ (Section 6.1–6.2).
Theorem 6.3 (Mod-8 Artin–Schreier)	[E]	Exact Modulo-8 identity $\bar{Y}_2 = \bar{x}^4\bar{x}_{\mathrm{AS}} + \bar{W}^2 + \bar{x}^2\bar{W}$ in $\mathbb{F}_2[t]$ forces $\deg Y \geq 4m_x + m_{\mathrm{AS}} + 1$ (and $\deg Y \geq 5m_x + 1$ for odd m_x); Modulo-16 invariance (Section 6.3).
Theorem 6.4 (Mod-9 & 2-Adic Laurent)	[E]	Modulo-9 Frobenius carry $2z_0z_1 + x_0^3x_1 \equiv \frac{1}{3}(Y_0^3 - z_0^2 - x_0^4 - 1) \pmod 3$ in $\mathbb{F}_3[t]$ and 2-adic Laurent integrality of $Y(t)^{3/2} - z(t)$ (Section 6.4).
Theorems 7.1–7.2 (General $d_x \geq 3$)	[E]	Conditional nonexistence for all $\deg x \geq 3$ under Strong Polynomial Hall–Vojta ($b_O = 0$) and Severi codimension obstructions (Section 7).

Table 1: Summary of theorems and verification status ([E] = complete elementary/analytic proof by hand; [C] = exact symbolic computation in SymPy).

1 Introduction and Context

Let $S \subset \mathbb{A}^3_{\mathbb{Q}}$ denote the smooth affine surface defined by

$$x^4 + y^3 + z^2 + 1 = 0 \quad \Longleftrightarrow \quad Y^3 - z^2 = x^4 + 1 \quad (Y = -y). \tag{1}$$

Whether $S(\mathbb{Z})$ is infinite is an open problem highlighted by Grechuk [1, 2]. Viewing S as the affine complement of the zero section O and the Kodaira type-IV fibre F_∞ on the minimal rational elliptic surface $\pi : \hat{E} \to \mathbb{P}^1_x$ with Weierstrass equation $z^2 = Y^3 - (x^4 + 1)$ and Mordell–Weil lattice $\mathrm{MW}(\hat{E}/\overline{\mathbb{Q}}(x)) \cong E_6^*$ [4, 5], the companion paper [3] proved via Frobenius trace computations at $p = 7$ that $\mathrm{MW}(\hat{E}/\mathbb{Q}(x)) = \{O\}$ and $\mathrm{Pic}(S_{\mathbb{Q}}) = 0$, establishing two foundational exclusions for polynomial families $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$:

- **Theorem A (**$\deg x \le 2$**):** There is no nonconstant polynomial solution $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ to (1) with $\deg x \le 2$ for any y, z.
- **Theorem B (**$\deg x = 3, \deg y \le 4$**):** There is no nonconstant polynomial solution in $\mathbb{Q}[t]^3$ with $\deg x = 3$ and $\deg y \le 4$ (which forces degrees $(3, 4, 6)$).

This report synthesizes and significantly extends the continuation programme into higher degrees $\deg x \ge 3$ (with a complete divisorial and syzygy theory for $\deg x = 4$ and universal bounds for all $\deg x \ge 3$) and into the arithmetic ring $\mathbb{Z}[t]^3$ (and $\mathbb{Z}_+[t]^3$), where we establish a hierarchy of unconditional 2-adic (Modulo-4 Frobenius, Modulo-8 Artin–Schreier, Modulo-16) and 3-adic (Modulo-3 differential, Modulo-9 Frobenius carry) obstructions alongside a universal Mutual Rational Root Locking Theorem over $\mathbb{Q}[t]$.

2 Universal Bounds, Divisorial Reduction, and Mutual Rational Root Locking for Arbitrary $\deg x \ge 1$

2.1 Positivity Lower Bound and Trace-Zero Divisorial Decomposition

Theorem 2.1 (Positivity Lower Bound and Trace-Zero Divisorial Reduction)**.** *Let $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ be a nonconstant solution to $x^4 + y^3 + z^2 + 1 = 0$, and put $Y(t) = -y(t)$, $d_x = \deg x \ge 1$, $d_Y = \deg Y$, and $d_z = \deg z$.*

(i) ***Universal lower bound over*** $\mathbb{R}$ ***and*** $\mathbb{Q}$***:*** *For every nonconstant solution in $\mathbb{R}[t]^3$ (and hence in $\mathbb{Q}[t]^3$), the leading terms of $x(t)^4$ and $z(t)^2$ have positive coefficients and cannot cancel, so $3d_Y = \max(4d_x, 2d_z) \ge 4d_x$. Consequently,*

$$\deg Y = d_Y \ge \left\lceil \frac{4d_x}{3} \right\rceil, \qquad \deg z = d_z \le \frac{3}{2} d_Y. \tag{2}$$

In particular, if $3 \nmid d_x$, then $3d_Y > 4d_x$, which forces $3d_Y = 2d_z$, so $d_Y = 2\nu$ and $d_z = 3\nu$ for some integer $\nu \ge \lceil (2d_x + 1)/3 \rceil$.

(ii) ***Galois trace and divisorial interpolation:*** *Assume $(x(t), Y(t), z(t))$ is injective on a dense open subset of $\mathbb{A}^1$. Let $t_1, \dots, t_{d_x}$ be the roots of $x(t) = x$ in $\overline{\mathbb{Q}(x)}$ and $P_i = (Y(t_i), z(t_i)) \in E(\overline{\mathbb{Q}(x)})$ the corresponding points on the generic fibre $E_x : z^2 = Y^3 - (x^4 + 1)$. Then*

$$P_1 + P_2 + \cdots + P_{d_x} = O \quad \text{in } E(\mathbb{Q}(x)) = \{O\}. \tag{3}$$

Consequently, there exist unique coprime polynomials $A(x, Y), B(x, Y) \in \mathbb{Q}[x, Y]$ with $\deg_Y A \le \lfloor d_x/2 \rfloor$ and $\deg_Y B \le \lfloor (d_x - 3)/2 \rfloor$ such that $A(x, Y) - B(x, Y)z$ vanishes on $\sum_{i=1}^{d_x} [P_i]$, and the characteristic polynomial $G_{d_x}(x, Y)$ of $Y(t)$ over $\mathbb{Q}(x)$ divides $A(x, Y)^2 - B(x, Y)^2(Y^3 - x^4 - 1)$ with quotient $H(x, Y) \in \mathbb{Q}[x, Y]$ of degree $\deg_Y H = \max(2 \deg_Y A, 2 \deg_Y B + 3) - d_x \in \{0, 1, 2\}$.

Proof. Part (i) follows immediately from $x_{d_x}^4 > 0$ and $z_{d_z}^2 \geq 0$ in $Y(t)^3 = x(t)^4 + z(t)^2 + 1$. For part (ii), the zero-cycle $\sum_{i=1}^{d_x}[P_i]$ on $E_{\mathbb{Q}(x)}$ is $\mathrm{Gal}(\overline{\mathbb{Q}(x)}/\mathbb{Q}(x))$-invariant, so its Abel–Jacobi sum lies in $E(\mathbb{Q}(x)) = \mathrm{MW}(\hat{E}/\mathbb{Q}(x)) = \{O\}$ by Proposition 2.3 of [3]. By Riemann–Roch on $E_{\mathbb{Q}(x)}$, $\mathcal{L}(d_x O)$ has dimension d_x with basis $\{1, Y, z, Y^2, Yz, \dots\}$ of pole orders $\{0, 2, 3, 4, 5, \dots, d_x\}$ at O. Since $\sum_{i=1}^{d_x}[P_i] \sim d_x O$, there is a unique function (up to $\mathbb{Q}(x)^*$-scaling) $f = A(x,Y) - B(x,Y)z \in \mathcal{L}(d_x O)$ vanishing on $P_1, \dots, P_{d_x}$, and taking norms from $\mathbb{Q}(x)[Y,z]/(z^2 - Y^3 + x^4 + 1)$ to $\mathbb{Q}(x)[Y]$ gives $A(x,Y)^2 - B(x,Y)^2(Y^3 - x^4 - 1) = H(x,Y)G_{d_x}(x,Y)$. □

2.2 Universal Upper Bounds for Arbitrary $\deg x \geq 3$

Theorem 2.2 (Universal Degree Bound for Arbitrary $\deg x \geq 3$)**.** *Let $(x(t), Y(t), z(t)) \in \mathbb{C}[t]^3$ be a nonconstant polynomial solution to $Y(t)^3 - z(t)^2 = x(t)^4 + 1$ with $d_x = \deg x \geq 3$, $d_Y = \deg Y$, and $d_z = \deg z$. Assume that $\gcd_{\mathbb{C}[t]}(Y(t), z(t)) = 1$ (which holds automatically over $\mathbb{Q}$ whenever every critical value of $x(t)$ has degree ≤ 3 over $\mathbb{Q}$, and in particular unconditionally over $\mathbb{Q}$ for $d_x \in \{3, 4\}$). Then (d_Y, d_z) falls into one of three finite regimes:*

(1) ***Sub-canonical regime*** *($3d_Y < 4d_x$**, over $\mathbb{C} \setminus \mathbb{R}$ only**): $d_z = 2d_x$ and $1 \leq d_Y \leq \lfloor (4d_x - 1)/3 \rfloor$. By Theorem 2.1(i), this regime contains **no solutions over $\mathbb{R}$ or $\mathbb{Q}$**.*

(2) ***Canonical regime*** *($3d_Y = 4d_x$**): $d_x = 3k$ is a multiple of 3, $d_Y = 4k$, and $d_z \leq 6k$ (with $d_z = 6k$ over $\mathbb{R}$ or $\mathbb{Q}$).*

(3) ***Hall–Davenport regime*** *($3d_Y = 2d_z > 4d_x$**): $d_Y = 2\nu$ is even and $d_z = 3\nu$ for an integer ν satisfying the universal bounds*

$$\left\lfloor \frac{2d_x}{3} \right\rfloor + 1 \leq \nu \leq 4d_x - 1, \qquad \text{so} \qquad \frac{4d_x + 2}{3} \leq d_Y \leq 8d_x - 2. \tag{4}$$

Without assuming $\gcd(Y(t), z(t)) = 1$, let $g_{\mathrm{cd}} = \deg \gcd(Y(t), z(t))$. Then $g_{\mathrm{cd}} \leq d_x - 1$, and the Mason–Stothers theorem gives $\nu \leq 5d_x - 2$ ($d_Y \leq 10d_x - 4$) unconditionally for all $d_x \geq 3$.

Proof. When $3d_Y > 4d_x$, the leading terms of $Y(t)^3$ and $z(t)^2$ must cancel, so $3d_Y = 2d_z \implies d_Y = 2\nu, d_z = 3\nu$ with $6\nu > 4d_x$. When $\gcd(Y, z) = 1$, Davenport's inequality [6, 7, 9] yields $4d_x = \deg(Y^3 - z^2) \geq \frac{1}{2}\deg Y + 1 = \nu + 1 \implies \nu \leq 4d_x - 1$. In general, any root α of $D_0 = \gcd(Y, z)$ is a root of $x(t)^4 + 1$ of multiplicity $m_\alpha \geq 2$, hence a root of $x'(t)$ of multiplicity $m_\alpha - 1 \geq \frac{1}{2}m_\alpha$, so $g_{\mathrm{cd}} \leq \deg x' = d_x - 1$; Mason–Stothers [8] applied to $(Y^3/D_0^2, -z^2/D_0^2, -(x^4+1)/D_0^2)$ yields $6\nu - 2g_{\mathrm{cd}} \leq 5\nu + 4d_x - g_{\mathrm{cd}} - 1 \implies \nu \leq 4d_x + g_{\mathrm{cd}} - 1 \leq 5d_x - 2$. For $d_x \in \{3, 4\}$ over $\mathbb{Q}$, any root α of $x'(t)$ has $[\mathbb{Q}(\alpha) : \mathbb{Q}] \leq d_x - 1 \leq 3$, whereas $X^4 + 1 = \Phi_8(X)$ is irreducible of degree 4 over $\mathbb{Q}$ and has no subfield of degree 3, so $x(\alpha)^4 + 1 \neq 0$ and $g_{\mathrm{cd}} = 0$. □

2.3 Mutual Rational Root Locking and Vanishing Multiplicity via $E_0(\mathbb{Q})$

We now establish a universal arithmetic constraint governing the rational and real roots of $x(t)$ and $z(t)$ for arbitrary degree $d_x \geq 1$.

Theorem 2.3 (Mutual Rational Root Locking and Even Multiplicity Theorem)**.** *Let $(x(t), Y(t), z(t)) \in \mathbb{Q}[t]^3$ be any polynomial solution to $Y(t)^3 - z(t)^2 = x(t)^4 + 1$ of arbitrary degree $d_x = \deg x \geq 1$.*

(i) ***Mutual rational root locking*** *($x(r) = 0 \iff z(r) = 0$**): For every rational number $r \in \mathbb{Q}$,***

$$x(r) = 0 \iff z(r) = 0 \quad \implies \quad Y(r) = 1. \tag{5}$$

In particular, $x(t)$ has a rational root $r \in \mathbb{Q}$ if and only if $z(t)$ has the exact same rational root $r \in \mathbb{Q}$.

(ii) ***Even vanishing multiplicity and stationary point of*** $Y(t)$*: At every rational root* $r \in \mathbb{Q}$ *of* $x(t)$ *(or* $z(t)$*), let* $e_x := \operatorname{ord}_{t=r}(x(t)) \geq 1$, $e_z := \operatorname{ord}_{t=r}(z(t)) \geq 1$, *and* $e_Y := \operatorname{ord}_{t=r}(Y(t)-1) \geq 1$. *Then*

$$e_Y = \min(4e_x, 2e_z) \geq 2 \quad \textit{is strictly even}, \qquad (t-r)^2 \mid (Y(t)-1), \qquad \textit{and} \qquad Y'(r) = 0. \tag{6}$$

More generally, at any real root $r \in \mathbb{R}$ *where* $x(r) = 0$, $Y(r) = (z(r)^2+1)^{1/3} \geq 1$ *and* $\operatorname{ord}_{t=r}(Y(t)-Y(r))$ *is even whenever* $z(r) = 0$.

(iii) ***Constraint on odd-*** z ***involutive families:*** *Suppose* $(x(t), Y(t), z(t)) \in \mathbb{Q}[t]^3$ *admits an involution* $\tilde{\sigma}(t) = -t$ *(after affine shift of* t*) under which* $Y(-t) = Y(t)$ *is even and* $z(-t) = -z(t)$ *is odd. Then* $z(0) = 0$ *with odd multiplicity* $e_z \equiv 1 \pmod 2$, *which forces* $x(0) = 0$, $Y(0) = 1$, *and* $2e_z \neq 4e_x$ *(since* $2e_z \equiv 2 \pmod 4$ *whereas* $4e_x \equiv 0 \pmod 4$*). In particular, if* $x(-t) = x(t)$ *is also even in* t, *then* $e_x \geq 2$ *is even,* $4e_x \geq 8$, *and for* $e_z = 1$ *we have* $e_Y = 2$ *with* $Y''(0)/2 = \frac{1}{3}z'(0)^2 > 0$.

Proof. (i) The Mordell curve $E_0 : w^2 = Y^3 - 1$ over $\mathbb{Q}$ (conductor 36, Cremona label `36a1`) has Mordell–Weil rank 0 and torsion subgroup $\mathbb{Z}/2\mathbb{Z}$ generated by $(1,0)$, so $E_0(\mathbb{Q}) = \{O, (1,0)\}$. If $r \in \mathbb{Q}$ satisfies $x(r) = 0$, evaluating $Y(t)^3 - z(t)^2 = x(t)^4 + 1$ at $t = r$ gives $z(r)^2 = Y(r)^3 - 1$, so $(Y(r), z(r)) \in E_0(\mathbb{Q}) = \{(1,0)\}$, whence $Y(r) = 1$ and $z(r) = 0$. Conversely, if $z(r) = 0$ with $r \in \mathbb{Q}$, evaluating at $t = r$ gives $(x(r)^2)^2 = Y(r)^3 - 1$, so $(Y(r), x(r)^2) \in E_0(\mathbb{Q}) = \{(1,0)\}$, whence $Y(r) = 1$ and $x(r)^2 = 0 \implies x(r) = 0$.

(ii) Write $x(t) = (t-r)^{e_x}\tilde{x}(t)$, $z(t) = (t-r)^{e_z}\tilde{z}(t)$, and $Y(t) = 1 + (t-r)^{e_Y}\tilde{Y}(t)$ with nonvanishing $\tilde{x}(r), \tilde{z}(r), \tilde{Y}(r) \in \mathbb{Q}^*$. Substituting into $Y(t)^3 - 1 = x(t)^4 + z(t)^2$ gives

$$(t-r)^{e_Y}\left(3\tilde{Y}(t) + 3(t-r)^{e_Y}\tilde{Y}(t)^2 + (t-r)^{2e_Y}\tilde{Y}(t)^3\right) = (t-r)^{4e_x}\tilde{x}(t)^4 + (t-r)^{2e_z}\tilde{z}(t)^2.$$

The left-hand side vanishes at $t = r$ to exact order e_Y since $3\tilde{Y}(r) \neq 0$. On the right-hand side, $\tilde{x}(r)^4 > 0$ and $\tilde{z}(r)^2 > 0$ are strictly positive rational numbers, so even when $4e_x = 2e_z$ their leading coefficient $\tilde{x}(r)^4 + \tilde{z}(r)^2 > 0$ cannot vanish. Thus $e_Y = \min(4e_x, 2e_z) \geq 2$ is even.

(iii) If $z(-t) = -z(t)$, then $z(0) = 0$ and $e_z = \operatorname{ord}_{t=0}(z(t))$ is odd. By (i), $x(0) = 0$ and $Y(0) = 1$, and by (ii), $e_Y = \min(4e_x, 2e_z)$ where $2e_z \equiv 2 \pmod 4 \neq 4e_x \equiv 0 \pmod 4$. □

Corollary 2.4 (Exclusion of Pullback and Unramified Families for Arbitrary $d_x \geq 3$)**.** *For any* $d_x = \deg x \geq 3$, *there is no nonconstant polynomial solution* $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ *to* (1) *satisfying any of the following:*

(a) $(x(t), y(t), z(t)) = (\tilde{x}(q(t)), \tilde{y}(q(t)), \tilde{z}(q(t)))$ *factors through* $q(t) \in \mathbb{Q}[t]$ *with* $\deg \tilde{x} \in \{1, 2\}$, *or with* $\deg \tilde{x} = 3$ *and* $\deg \tilde{y} \leq 10$;

(b) $Y(t) = Y_0(x(t))$ *lies in* $\mathbb{Q}[x(t)]$;

(c) $z(t) = z_0(x(t))$ *lies in* $\mathbb{Q}[x(t)]$.

3 Extension to $\deg x = 4$: Quartic Divisorial Reduction and Coupled Syzygy System

For $d_x = \deg x = 4$, Theorem 2.2 confines any rational polynomial solution to the Hall–Davenport regime $\deg Y = 2\nu \in \{6, 8, \ldots, 30\}$ $(3 \leq \nu \leq 15)$ with $\deg z = 3\nu$. We now prove that for $\deg x = 4$, UFD coprimality in $\mathbb{Q}[x, Y]$ forces the Y^2-coefficient $q(x)$ in the Riemann–Roch interpolant to be a nonzero constant, yielding an explicit coupled syzygy system in $\mathbb{Q}[x]$.

Theorem 3.1 (Quartic Divisorial Reduction and Coupled Syzygy System for $\deg x = 4$)**.** *Let $(x(t), Y(t), z(t)) \in \mathbb{Q}[t]^3$ be a polynomial solution to (1) with $\deg x = 4$ and $\deg Y = 2\nu$ ($3 \le \nu \le 15$) such that $[\mathbb{Q}(t) : \mathbb{Q}(x, Y)] = 1$ (so the characteristic polynomial $G_4(x, Y) = Y^4 - e_1(x)Y^3 + e_2(x)Y^2 - e_3(x)Y + e_4(x) \in \mathbb{Q}[x][Y]$ of $Y(t)$ over $\mathbb{Q}(x)$ is irreducible, with $x(t)$ normalised to be monic in t).*

(i) ***Constancy of the Y^2-interpolant coefficient*** *$q(x)$: In the Riemann–Roch relation $B(x)z = q(x)Y^2 + m(x)Y + n(x)$ on $E_{\mathbb{Q}(x)}$ with $\gcd_{\mathbb{Q}[x]}(B, q, m, n) = 1$, the polynomial $q(x) = q_0 \in \mathbb{Q}^*$ is a* ***nonzero constant****. Scaling (B, m, n) by q_0^{-1} so that $q_0 = 1$, we have the exact identity in $\mathbb{Q}[x, Y]$:*

$$\big(Y^2 + m(x)Y + n(x)\big)^2 - B(x)^2\left(Y^3 - x^4 - 1\right) = Y^4 - e_1(x)Y^3 + e_2(x)Y^2 - e_3(x)Y + e_4(x), \tag{7}$$

so that the elementary symmetric polynomials $e_1, e_2, e_3, e_4 \in \mathbb{Q}[x]$ of $Y(t_1), Y(t_2), Y(t_3), Y(t_4)$ are given by

$$\begin{aligned} e_1(x) &= B(x)^2 - 2m(x), & e_2(x) &= m(x)^2 + 2n(x), \\ e_3(x) &= -2m(x)n(x), & e_4(x) &= n(x)^2 + B(x)^2(x^4+1). \end{aligned} \tag{8}$$

(ii) ***Universal coupled syzygy system in $\mathbb{Q}[x]$:*** *Eliminating $B(x)^2 = e_1(x) + 2m(x)$ and $2n(x) = e_2(x) - m(x)^2$ from (8) yields the coupled polynomial identities in $\mathbb{Q}[x]$:*

$$\Phi_{4,1}(x) := m(x)^3 - e_2(x)m(x) - e_3(x) \equiv 0, \tag{9}$$

$$\Phi_{4,2}(x) := \big(e_2(x) - m(x)^2\big)^2 + 4\left(e_1(x) + 2m(x)\right)(x^4+1) - 4e_4(x) \equiv 0, \tag{10}$$

subject to the square condition $e_1(x) + 2m(x) = B(x)^2$ in $\mathbb{Q}[x]$ and divisibility $B(x(t)) \mid (Y(t)^2 + m(x(t))Y(t) + n(x(t)))$ in $\mathbb{Q}[t]$. In particular, $m(x)$ divides $e_3(x)$ in $\mathbb{Q}[x]$.

(iii) ***Exact degree bounds and leading-term structure:*** *For every $\nu \in \{3, 4, \dots, 15\}$, the polynomials $B(x), m(x), n(x) \in \mathbb{Q}[x]$ satisfy the exact degree bounds*

$$\deg_x B(x) \le \left\lfloor \frac{\nu}{4} \right\rfloor, \qquad \deg_x m(x) \le \left\lfloor \frac{\nu}{2} \right\rfloor, \qquad \deg_x n(x) = \nu. \tag{11}$$

In particular, for $\nu = 3$ ($\deg x = 4, \deg Y = 6, \deg z = 9$), $B(x) = B_0 \in \mathbb{Q}^*$ is ***constant****, so $z(t) = B_0^{-1}\left(Y(t)^2 + m(x(t))Y(t) + n(x(t))\right)$ is a polynomial in $(x(t), Y(t))$ with $\deg_x m \le 1$ and $\deg_x n = 3$. Furthermore, for every odd ν, the leading coefficient $Y_{2\nu} = \mathrm{LC}(Y) > 0$ forces $\mathrm{LC}(e_2) = -2Y_{2\nu}^2 < 0$ and $\mathrm{LC}(n) = -Y_{2\nu}^2 < 0$, cancelling the $t^{4\nu}$ leading term of $Y(t)^2 + n(x(t))$.*

Proof. (i) Since $x(t)$ is monic of degree 4, $\mathbb{Q}[t]$ is a free $\mathbb{Q}[x]$-module of rank 4, so $Y(t) \in \mathbb{Q}[t]$ is integral over $\mathbb{Q}[x]$ and its minimal polynomial $G_4(x, Y)$ is monic in Y with coefficients $e_j(x) \in \mathbb{Q}[x]$. In $\mathcal{L}(4O) = \langle 1, Y, z, Y^2\rangle$, the relation $B(x)z = q(x)Y^2 + m(x)Y + n(x)$ has $B(x) \neq 0$ (as $\deg_Y G_4 = 4 > 2$) and $q(x) \neq 0$ (as a line $B(x)z = m(x)Y + n(x)$ meets the cubic E_x in at most 3 points). Since $(q(x)Y^2 + m(x)Y + n(x))^2 - B(x)^2(Y^3 - x^4 - 1)$ vanishes at all 4 roots of the irreducible monic quartic $G_4(x, Y)$ and has Y^4-coefficient $q(x)^2$, it equals $q(x)^2 G_4(x, Y)$ in $\mathbb{Q}[x, Y]$:

$$(qY^2 + mY + n)^2 - B^2(Y^3 - x^4 - 1) = q^2(Y^4 - e_1Y^3 + e_2Y^2 - e_3Y + e_4).$$

Let $\pi(x) \in \mathbb{Q}[x]$ be any irreducible factor of $q(x)$. Comparing Y^3, Y^2, and Y^0 gives $2qm - B^2 = -q^2e_1$, $m^2 + 2qn = q^2e_2$, and $n^2 + B^2(x^4+1) = q^2e_4$. Thus $\pi \mid q \implies \pi \mid B^2 \implies \pi \mid B$; similarly $\pi \mid q \implies \pi \mid m^2 \implies \pi \mid m$; and since $\pi^2 \mid q^2e_4$ and $\pi^2 \mid B^2(x^4+1)$, we obtain $\pi^2 \mid n^2 \implies \pi \mid n$. This contradicts $\gcd_{\mathbb{Q}[x]}(B, q, m, n) = 1$, proving $q(x) = q_0 \in \mathbb{Q}^*$. Scaling by q_0^{-1} gives (7)–(8).

(ii) From $e_2 = m^2 + 2n$, we have $2n = e_2 - m^2$. Substituting into $e_3 = -2mn = -m(e_2 - m^2)$ gives (9), and substituting $B^2 = e_1 + 2m$ and $2n = e_2 - m^2$ into $4e_4 = (2n)^2 + 4B^2(x^4+1)$ gives (10).

(iii) Since $\deg_t Y(t) = 2\nu$ and $\deg_t x(t) = 4$, the symmetric functions $e_k(x)$ have $\deg_x e_k \le \lfloor 2k\nu/4 \rfloor = \lfloor k\nu/2 \rfloor$, with $\deg_x e_4 = 2\nu$. In $e_4(x) = n(x)^2 + B(x)^2(x^4+1)$, both $n(x)^2$ and $B(x)^2x^4$ have positive leading coefficients over $\mathbb{R}$, so $\deg_x n \le \nu$ and $\deg_x B \le \nu - 2$. Then $m(x)^2 = e_2(x) - 2n(x)$ has degree $\le \nu$, so $\deg_x m \le \lfloor \nu/2 \rfloor$, and $B(x)^2 = e_1(x) + 2m(x)$ has degree $\le \lfloor \nu/2 \rfloor$, forcing $\deg_x B \le \lfloor \nu/4 \rfloor$. Since $2\lfloor \nu/4 \rfloor + 4 \le \frac{\nu}{2} + 4 < 2\nu$ for all $\nu \ge 3$, $\deg_x(n^2) = 2\nu \implies \deg_x n = \nu$. Finally, when ν is odd, the 4 roots $t_k = i^k x^{1/4} + O(1)$ give $Y(t_k) = (-1)^k Y_{2\nu} x^{\nu/2} + O(x^{(\nu-1)/2})$, so $\prod_{k=0}^{3}(Y - Y(t_k)) = (Y^2 - Y_{2\nu}^2 x^\nu)^2 + \ldots$, giving $\mathrm{LC}(e_2) = -2Y_{2\nu}^2$, and since $\deg_x(m^2) \le \nu - 1 < \nu$, $2\,\mathrm{LC}(n) = \mathrm{LC}(e_2) = -2Y_{2\nu}^2 \implies \mathrm{LC}(n) = -Y_{2\nu}^2$. □

4 Binary Cubic Covariants and Complete Resolution of $\nu = 4$ ($\deg Y = 8$) and $\nu = 5$ ($\deg Y = 10$) for $\deg x = 3$

4.1 Covariant Structure and Unified Bihomogeneous Identity for $\deg x = 3$

For $\deg x = 3$ and $\deg Y = 2\nu \in \{8, 10, 14, 16, 20, 22\}$, normalise $x(t) = t^3 + st + x_0$ and put $u = x - x_0 = t^3 + st$. By Lemma 6.1 of [3], $z(t) = m(u)Y(t) + n(u)$ with $m(u), n(u) \in \mathbb{Q}[u]$, $m(u) \neq 0$, and $Y(t) = \frac{1}{3}m(u)^2 + W(t)$, where $W(t) = \frac{2}{3}s\gamma(u) + \beta(u)t + \gamma(u)t^2$ has $\mathrm{Tr}_{\mathbb{Q}(t)/\mathbb{Q}(u)}(W) = 0$.

Theorem 4.1 (Binary Cubic Covariant Structure and Unified Bihomogeneous Identity)**.** *Let $W(t) = \frac{2}{3}s\gamma(u) + \beta(u)t + \gamma(u)t^2 \in \mathbb{Q}[u][t]/(t^3 + st - u)$, and let $T^3 + p(u)T - N(u)$ be the characteristic polynomial of W over $\mathbb{Q}[u]$.*

(i) ***Covariant representation and discriminant syzygy:*** *Let $I(\beta, \gamma) = \beta^3 + s\beta\gamma^2 - u\gamma^3$ be the index cubic of W. Then $p(u)$ and $N(u)$ are the Hessian and Jacobian covariants of $I(\beta, \gamma)$:*

$$p(u) = \frac{1}{12}\,\mathrm{Hess}_{\beta,\gamma}(I) = s\beta^2 - 3u\beta\gamma - \frac{1}{3}s^2\gamma^2, \tag{12}$$

$$N(u) = -\frac{1}{9}\,\mathrm{Jac}_{\beta,\gamma}(I, p) = u\beta^3 + \frac{2}{3}s^2\beta^2\gamma - su\beta\gamma^2 + \left(u^2 + \frac{2}{27}s^3\right)\gamma^3, \tag{13}$$

satisfying the universal covariant syzygy $4p(u)^3 + 27N(u)^2 = (27u^2 + 4s^3)I(\beta(u), \gamma(u))^2$.

(ii) ***Unified polynomial identity:*** *A tuple $(s, x_0, \beta(u), \gamma(u), m(u))$ with $m(u) \neq 0$ defines a polynomial solution to (1) if and only if it satisfies the single polynomial identity in $\mathbb{Q}[u]$:*

$$\Phi(u) := \Psi(u) - 4m(u)^2\left((u + x_0)^4 + 1\right) \equiv 0, \qquad \Psi(u) := 4m^2N - p^2 + \frac{2}{3}m^4p + \frac{1}{27}m^8, \tag{14}$$

which automatically implies $m(u) \mid p(u)$ and $n(u) = -\frac{p(u)}{2m(u)} - \frac{1}{6}m(u)^3 \in \mathbb{Q}[u]$.

(iii) ***Exact degree bounds and $\mathbb{G}_m^2$-bihomogeneity:*** *For $d_Y = 2\nu \in \{8, 10, 14, 16, 20, 22\}$,*

$$\deg_u \beta \le \left\lfloor \frac{2\nu - 1}{3} \right\rfloor, \quad \deg_u \gamma \le \left\lfloor \frac{2\nu - 2}{3} \right\rfloor, \quad \mu := \deg_u m \le \left\lfloor \frac{\nu}{3} \right\rfloor \le \frac{\nu - 1}{3}, \quad \deg_u n = \nu, \tag{15}$$

and $\Psi(u)$ is bihomogeneous of weight μ^8 under the torus $\mathbb{G}_m^2 = \{(\lambda, \mu) \in (\overline{\mathbb{Q}}^*)^2\}$ acting via

$$(u, s, x_0) \mapsto (\lambda^3 u, \lambda^2 s, \lambda^3 x_0), \qquad (\beta, \gamma, m, n) \mapsto (\lambda^{-1}\mu^2\beta, \lambda^{-2}\mu^2\gamma, \mu m, \mu^3 n). \tag{16}$$

4.2 Complete Resolution of the $\nu = 4$ ($\deg x = 3, \deg Y = 8, \deg z = 12$) System

For $\nu = 4$, writing $\beta(u) = b_2u^2 + b_1u + b_0$, $\gamma(u) = g_2u^2 + g_1u + g_0$ ($g_2 \neq 0$), and $m(u) = m_1u + m_0$, the leading coefficient of $\Phi(u)$ at u^{10} is $\mathrm{coeff}_{u^{10}}(\Phi) = -g_2^2(9b_2^2 - 4g_2m_1^2) = 0 \implies 9b_2^2 = 4g_2m_1^2$.

Theorem 4.2 (Complete Solution of the $m_1 = 0$ Branch for $\nu = 4$)**.** *In the case* $\deg x = 3, \deg Y = 8, \deg z = 12$ *with* $m(u) = m_0 \in \overline{\mathbb{Q}}^*$ *constant (so* $b_2 = 0$*):*

(i) ***No rational solutions:*** *There is no solution* $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ *with* $m_1 = 0$*.*

(ii) ***Explicit solution over*** $\mathbb{Q}(\sqrt{3}, 2^{1/3})$***:*** *Over* $\overline{\mathbb{Q}}$*, setting* $s = 0, x_0 = 0, b_1 = -2, g_2 = \frac{3}{2}2^{1/3}, m_0 = -2^{1/3}\sqrt{3}$ *yields the explicit polynomial identity of degrees* $(3, 8, 12)$*:*

$$x(t) = t^3, \quad Y(t) = \frac{3}{2}2^{1/3}t^8 - 2t^4 + 2^{2/3}, \quad z(t) = \sqrt{3}\left(\frac{3}{2}t^{12} - \frac{3}{2}2^{2/3}t^8 + 2\cdot 2^{1/3}t^4 - 1\right), \tag{17}$$

which satisfies $Y(t)^3 - z(t)^2 = x(t)^4 + 1$ *identically in* $\mathbb{Q}(\sqrt{3}, 2^{1/3})[t]$*, disproving the heuristic expectation in Appendix C of [3] that the* $\nu \geq 4$ *moduli spaces are empty over* $\overline{\mathbb{Q}}$*.*

Theorem 4.3 (Nonexistence of Solutions in the $m_1 \neq 0$ Branch for $\nu = 4$)**.** *There is no polynomial solution* $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ *to* (1) *with* $\deg x = 3$ *and* $\deg y = 8$ *(*$\deg z = 12$*).*

Summary of Proof (Theorems 4.2–4.3). For $m_1 = 0$, u^8, u^7, u^6 solve linearly for m_0^2, b_0, g_0, and u^5 factors into $s = 0$ (forcing $b_1^3 = -8, g_1 = x_0 = 0, g_2^3 = 27/4 \notin (\mathbb{Q}^*)^3$, yielding (17)), $g_1 = 0, s \neq 0$ (contradiction), and $g_1 s \neq 0$ (reducing to coprime univariate polynomials $P_2(\lambda), P_1(\lambda)$ of degrees 7 and 9 with $\gcd(P_1, P_2) = 1$). For $m_1 \neq 0$, scaling $b_2 = 1, m_1 = 1, g_2 = 9/4$ under (16) solves u^9, u^8, u^7 linearly for m_0, b_0, g_0 with monomial denominator $\tilde{D}$. Under the blow-up $E = \tilde{D}f/S$, $p(-m_0) = 0$ reduces to the quadratic $P_{\text{rem}}(\tilde{D}, S, f) = A_0\tilde{D}^2 + B_0\tilde{D} + C_0 = 0$, and pseudo-dividing the fourth-power invariants I_2, I_3 by P_{rem} eliminates $\tilde{D}$ and $S = W_S(f + 12)$ to a degree-530 resultant $R_{12,23}(f) = c \cdot f^{80}(f+6)^{40}(f+12)^{24}(f-12)^{20}(f^2+18f+108)^{16}(f+9) \cdot H_{\text{sp}}(f)^2 \cdot H_{\text{true}}(f) \in \mathbb{Z}[f]$, whose true degree-117 factor $H_{\text{true}}(f)$ has irreducible factor degrees $[2, 8, 9, 10, 22, 25, 41]$ mod 10037, $[2, 3, 6, 18, 19, 69]$ mod 10067, and $[2, 2, 4, 5, 18, 86]$ mod 10133, hence no rational root $f \in \mathbb{Q}$. □

4.3 Complete Resolution of the $\nu = 5$ ($\deg x = 3, \deg Y = 10, \deg z = 15$) System

For $\nu = 5$ ($d_Y = 10, d_z = 15$), $\deg_u \beta = 3$ ($b_3 \neq 0$), $\deg_u \gamma \leq 2$, and $\deg_u m \leq 1$. Since $\deg_u(m(u)^2((u+x_0)^4+1)) = 2\mu + 4 \leq 6$, all coefficients of u^k in $\Phi(u)$ for $k \geq 7$ depend purely on the bihomogeneous polynomial $\Psi(u)$.

Theorem 4.4 (Exclusion of the Constant-m Branch for $\nu = 5$)**.** *There is no polynomial solution* $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ *to* (1) *with* $\deg x = 3, \deg y = 10, \deg z = 15$*, and* $m(u) = m_0 \in \mathbb{Q}^*$ *constant. In fact, no complex polynomial solution in* $\mathbb{C}[t]^3$ *with* $m(u) \in \mathbb{C}^*$ *constant satisfies* $\deg_t(Y^3 - z^2) \leq 12$*.*

Proof. Setting $b_3 = 1$, $\deg_u p \leq 5$ forces $g_2 = s/3$, and u^{10}, u^9, u^8, u^7 of $\Psi(u)$ solve linearly for $m_0^2 = \frac{1}{4}(b_2 s - 3g_1)^2, b_1, b_0, g_0$. Substituting into u^6 and u^5 with $b_2 = \lambda g_1/s$ and $s^5 = \mu g_1^2$ yields two bivariate polynomials $P_6(\lambda, \mu)$ (cubic in μ) and $P_5(\lambda, \mu)$ (quartic in μ) with resultant

$$\begin{aligned}\text{Res}_\mu(P_6, P_5) = {} & 88529418169417209544704\,\lambda^{12}(\lambda-3)^5 \\ & \times \left(113\lambda^5 - 5181\lambda^4 + 94500\lambda^3 - 830250\lambda^2 + 3290625\lambda - 3796875\right).\end{aligned}$$

Here $\lambda \notin \{0, 3\}$, and the quintic factor $Q_5(\lambda)$ is irreducible over $\mathbb{Q}$ (modulo 7, $Q_5(\lambda) \equiv \lambda^5 + \lambda^4 + 3\lambda^2 + 4\lambda + 1$ is irreducible) and shares no complex root with $u^4, \ldots, u^1$. □

Theorem 4.5 (Denominator-Free Reduction and Euclidean Elimination of the Linear-m Branch for $\nu = 5$)**.** *In the linear-m branch* $m(u) = m_1 u + m_0$ *(*$m_1 \neq 0$*) for* $\nu = 5$ *(*$\deg x = 3, \deg Y = 10, \deg z = 15$*), the* $\mathbb{G}_m^2$*-normalised system (*$b_3 = 1, m_1 = 1, g_2 = (s-2)/3$*) satisfies:*

(i) ***Denominator-free coordinate parametrization and involution symmetry:*** *Introducing the polynomial coordinates* $(B_2, F_1, \tilde{B}_1)$ *defined by*

$$\begin{aligned} b_2 &= (s-1)B_2, \qquad g_1 = \frac{(s-1)\,(F_1 + B_2(s-3))}{6}, \\ b_1 &= \frac{1}{4}(s-1)^2 B_2^2 + \frac{2(s-1)\left((s+1)^2 + \tilde{B}_1\right)}{27}, \end{aligned} \tag{18}$$

solving u^{11}, u^{10}, u^9 *for* m_0, g_0, b_0 *cancels the denominator* $(s-1)$ *identically from* b_0*, making* $\beta(u), \gamma(u), m(u),\ p(u), N(u)$*, and* $\Psi(u)$ ***pure polynomials in*** $\mathbb{Q}[u, B_2, F_1, \tilde{B}_1, s]$ ***with zero denominators****. Moreover,* $\Psi(-u, -B_2, -F_1, \tilde{B}_1, s) = +\Psi(u, B_2, F_1, \tilde{B}_1, s)$*, so every coefficient* $\Psi_k = \mathrm{coeff}_{u^k}(\Psi)$ *has exact parity* $(-1)^k$ *in* (B_2, F_1)*.*

(ii) ***Exact rational elimination of*** $Z = (s-1)^4 B_2^2$ ***and*** $L = \lambda + s$***:*** *Setting* $F_1 = (L-s)B_2$*,* $\Psi_8 = 0$ *is linear in* $Z = \tilde{b}_2^2 = (s-1)^4 B_2^2$*:*

$$Z = \tilde{b}_2^2 = -\frac{8\tilde{B}_1(s-1)^2\left(-\tilde{B}_1 + 2s(s^2-1)\right)}{27\left(\tilde{B}_1(L-s)^2 - \tilde{B}_1 + 2L^2\right)}. \tag{19}$$

Substituting (19) *into* $\Psi_7/\tilde{b}_2 = 0$ *and* $p(-m_0) = 0$ *yields polynomials* $\mathrm{eq}_{7,L}(\tilde{B}_1, L, s)$ *(degree* 3 *in* L*) and* $\mathrm{eq}_{p,L}(\tilde{B}_1, L, s)$ *(degree* 4 *in* L*). Two-step Euclidean pseudo-division in* L *solves rationally for* L*:*

$$L = -\frac{A_0(\tilde{B}_1, s)}{A_1(\tilde{B}_1, s)} \in \mathbb{Q}(\tilde{B}_1, s), \tag{20}$$

where $A_1, A_0 \in \mathbb{Z}[\tilde{B}_1, s]$ *have degrees* $(10, 22)$ *and* $(10, 24)$*, and reduces the resultant of* $\mathrm{eq}_{7,L}$ *and* $\mathrm{eq}_{p,L}$ *with respect to* L *to the irreducible plane curve* $H_9(\tilde{B}_1, s) = 256s^2(3s-1)\tilde{B}_1^9 - \cdots + (125s^{12} + \cdots - 4)^2 = 0$ *of degree* $(9, 24)$*.*

(iii) ***Exact*** Z***-degree drop and overdetermination by*** $I_2 = I_3 = 0$***:*** *In the exact quotient* $Q_{\mathrm{quot}}(u) = \Psi(u)/(4(u+m_0)^2) = Q_4u^4 + Q_3u^3 + Q_2u^2 + Q_1u + Q_0$*, the* Z^3 *term in* $4Q_3 = \Psi_5 - 2m_0\Psi_6$ *cancels identically, so* Q_4, Q_3 *have degree* 2 *in* Z *and* Q_2, Q_1 *have degree* 3 *in* Z*. Substituting* Z *and* $L = -A_0/A_1$ *into* $I_2 = 3Q_3^2 - 8Q_4Q_2 = 0$ *and* $I_3 = Q_3Q_2 - 6Q_4Q_1 = 0$ *yields two independent bivariate equations* $G_{I_2}(\tilde{B}_1, s) = 0$ *and* $G_{I_3}(\tilde{B}_1, s) = 0$ *of reduced degrees* $(47, 95)$ *and* $(50, 101)$ *overdetermining* $H_9(\tilde{B}_1, s) = 0$*.*

5 Pullback Elliptic Surfaces and Infinite Towers over $\overline{\mathbb{Q}}$ for $\deg x = 3k$

Theorem 5.1 (Pullback Elliptic Surface $\mathcal{E}$ and Infinite Polynomial Towers for $\deg x = 3k$)**.** *Let* $v(t) \in \mathbb{Q}[t]$ *be any nonconstant polynomial of degree* $k \geq 1$*, and set* $x(t) = v(t)^3$ *(*$\deg x = 3k$*) and* $w(t) = v(t)^4$ *(*$\deg w = 4k$*). Under* $x(t)^4 + 1 = w(t)^3 + 1$*, the surface* S *pulls back to the minimal rational elliptic surface* $\mathcal{E} : \tilde{z}^2 = \tilde{Y}^3 - (w^3+1)$ *over* $\mathbb{P}^1_w$ *with singular fibres* $3 \times \mathrm{II} + \mathrm{I}_0^*$ *and Mordell–Weil lattice* $\mathrm{MW}(\mathcal{E}/\overline{\mathbb{Q}}(w)) \cong D_4^*$ *of rank* 4*. Evaluating the explicit* $\mathbb{Q}(\sqrt{3}, 2^{1/3})$*-rational sections*

$$s_1(w) = \left(w + 2^{2/3},\ \sqrt{3}\left(2^{1/3}w + 1\right)\right), \tag{21}$$

$$s_2(w) = \left(\frac{3}{2}2^{1/3}w^2 - 2w + 2^{2/3},\ \sqrt{3}\left(\frac{3}{2}w^3 - \frac{3}{2}2^{2/3}w^2 + 2\cdot 2^{1/3}w - 1\right)\right) \tag{22}$$

at $w = v(t)^4$ *yields infinite families of polynomial identities over* $\mathbb{Q}(\sqrt{3}, 2^{1/3})$ *of degrees* $(3k, 4k, 6k)$ *and* $(3k, 8k, 12k)$ *for every* $k \geq 1$*, whereas* $\mathrm{MW}(\mathcal{E}/\mathbb{Q}(w)) = \{O\}$ *obstructs descent to* $\mathbb{Q}$*.*

6 Unconditional Modulo-4 Frobenius, Modulo-8 Artin–Schreier, Modulo-9 Carry, and 2-Adic Obstructions in $\mathbb{Z}[t]$

We now turn to the question of polynomial solutions $(x(t), Y(t), z(t)) \in \mathbb{Z}[t]^3$ (and $\mathbb{Z}_+[t]^3$) to $x^4 - Y^3 + z^2 + 1 = 0$ over the integers.

6.1 Positivity Shift Equivalence and Modulo-4 Frobenius Obstruction

Theorem 6.1 (Positivity Shift Equivalence $\mathbb{Z}[t]^3 \iff \mathbb{Z}_+[t]^3$)**.** *For any degree triple (d_x, d_Y, d_z) with $d_x \geq 1$, a nonconstant polynomial solution $(x(t), Y(t), z(t)) \in \mathbb{Z}[t]^3$ of degrees (d_x, d_Y, d_z) to $x^4 - Y^3 + z^2 + 1 = 0$ exists if and only if there exists a polynomial solution $(\tilde{x}(t), \tilde{Y}(t), \tilde{z}(t)) \in \mathbb{Z}_+[t]^3$ of degrees (d_x, d_Y, d_z) all of whose coefficients are strictly positive integers.*

Proof. Replacing $x(t) \mapsto \pm x(t)$ and $z(t) \mapsto \pm z(t)$ ensures $\mathrm{LC}(x) \geq 1$ and $\mathrm{LC}(z) \geq 1$, which forces $\mathrm{LC}(Y) \geq 1$ from $Y^3 = x^4 + z^2 + 1$. For any $P(t) \in \mathbb{Z}[t]$ with $\mathrm{LC}(P) \geq 1$, $P^{(k)}(c) > 0$ for all $0 \leq k \leq \deg P$ and all sufficiently large $c \in \mathbb{Z}_+$. Thus $(\tilde{x}(t), \tilde{Y}(t), \tilde{z}(t)) = (x(t+c), Y(t+c), z(t+c)) \in \mathbb{Z}_+[t]^3$ has all Taylor coefficients $\frac{1}{k!}P^{(k)}(c) \in \mathbb{Z}_+$. □

Theorem 6.2 (The Modulo-4 Frobenius, $\mathbb{F}_4$-Root, and $\mathbb{F}_2[t]$-Derivative Obstruction in $\mathbb{Z}[t]$)**.** *Let $(x(t), Y(t), z(t)) \in \mathbb{Z}[t]^3$ be any polynomial solution to $x(t)^4 - Y(t)^3 + z(t)^2 + 1 = 0$, and denote by $\bar{f}(t) := f(t) \bmod 2 \in \mathbb{F}_2[t]$ the reduction modulo 2.*

(i) ***Modulo-2 fourth-power law and exact modulo-4 syzygy:*** *There exist polynomials $w(t), q(t), v(t) \in \mathbb{Z}[t]$ with $\gcd_{\mathbb{F}_2[t]}(\bar{w}, \bar{q}) = 1$ such that in $\mathbb{F}_2[t]$:*

$$\begin{aligned} \bar{Y}(t) &= \bar{w}(t)^4, & \bar{z}(t) &= \bar{w}(t)^6 + \bar{x}(t)^2 + 1, \\ \bar{w}(t)^3 \bar{q}(t) &= \bar{x}(t)^2 + \bar{x}(t) + 1, & \bar{q}(t) &= \bar{w}(t)\bar{v}(t) + \bar{x}(t) + 1. \end{aligned} \tag{23}$$

(ii) ***Even-degree irreducible factor law ($\mathbb{F}_4$-obstruction):*** *Every irreducible factor $\pi(t) \in \mathbb{F}_2[t]$ of $\bar{w}(t)$ divides $\bar{x}(t)^2 + \bar{x}(t) + 1$, so every root $\alpha \in \overline{\mathbb{F}_2}$ of $\bar{w}(t)$ satisfies $\bar{x}(\alpha) \in \mathbb{F}_4 \setminus \mathbb{F}_2$ and $\mathbb{F}_4 \subseteq \mathbb{F}_2(\alpha)$. Consequently, every irreducible factor of $\bar{w}(t)$ has even degree over $\mathbb{F}_2$, $\deg_{\mathbb{F}_2} \bar{w}(t)$ is even, and*

$$\deg_{\mathbb{F}_2}(Y(t) \bmod 2) = 4 \deg_{\mathbb{F}_2} \bar{w}(t) \equiv 0 \pmod 8. \tag{24}$$

(iii) ***Formal derivative divisibility and degree bound in $\mathbb{F}_2[t]$:*** *Differentiating $\bar{w}^3\bar{q} = \bar{x}^2 + \bar{x} + 1$ in $\mathbb{F}_2[t]$ yields $\bar{w}(t)^2(\bar{w}'\bar{q} + \bar{w}\bar{q}') = \bar{x}'(t)$, so $\bar{w}(t)^2 \mid \bar{x}'(t)$ in $\mathbb{F}_2[t]$. By Frobenius 2^r-descent on $\bar{x}(t)$, if $\bar{Y}(t) \neq 1$ in $\mathbb{F}_2[t]$ (i.e., $\deg_{\mathbb{F}_2} \bar{w} \geq 2$), then $\deg_{\mathbb{F}_2} \bar{x} \geq 5$ and*

$$8 \leq \deg_{\mathbb{F}_2}(Y(t) \bmod 2) \leq 2 \deg_{\mathbb{F}_2}(x(t) \bmod 2) - 2 \leq 2 \deg x - 2. \tag{25}$$

(iv) ***Universal modulo-4 collapse for $\deg x \leq 4$:*** *Whenever $\bar{w}(t) = 1$ (which holds unconditionally for every solution in $\mathbb{Z}[t]^3$ with $\deg_{\mathbb{F}_2}(x \bmod 2) \leq 4$, and in particular for all $\deg x \leq 4$), we have in $(\mathbb{Z}/4\mathbb{Z})[t]$:*

$$Y(t) \equiv 1 + 2x(t)^4 \pmod 4, \quad z(t) \equiv x(t)^2 \pmod 2, \quad \deg Y(t) \geq 4m_x = 4 \deg_{\mathbb{F}_2}(x \bmod 2). \tag{26}$$

6.2 The Modulo-8 Artin–Schreier and Modulo-16 2-Adic Degree Jump Theorem in $\mathbb{Z}[t]$

We now push the 2-adic expansion of Theorem 6.2(iv) to Modulo 8 and Modulo 16, uncovering an exact Artin–Schreier operator structure in $\mathbb{F}_2[t]$ that forces a sharp jump in $\deg Y(t)$ beyond $4m_x$.

Theorem 6.3 (Modulo-8 Artin–Schreier and Modulo-16 2-Adic Degree Jump Theorem in $\mathbb{Z}[t]$)**.** *Let $(x(t),Y(t),z(t))\in\mathbb{Z}[t]^3$ be any polynomial solution to $x(t)^4-Y(t)^3+z(t)^2+1=0$ with $\bar{w}(t)=1$ in $\mathbb{F}_2[t]$ (which holds unconditionally for all $\deg_{\mathbb{F}_2}(x \bmod 2)\le 4$, and whenever $8\nmid \deg_{\mathbb{F}_2}(Y \bmod 2)$). By Theorem 6.2(iv), write*

$$Y(t)=1+2x(t)^4+4Y_2(t),\qquad z(t)=x(t)^2+2Z_1(t),\qquad \text{with } Y_2(t),Z_1(t)\in\mathbb{Z}[t]. \tag{27}$$

(i) ***Exact identity in $\mathbb{Z}[t]$ and Modulo-8 Artin–Schreier reduction in $\mathbb{F}_2[t]$:*** *Substituting (27) into $(Y(t)^3-z(t)^2-x(t)^4-1)/4=0$ yields the exact polynomial identity in $\mathbb{Z}[t]$:*

$$Z_1^2+x^2Z_1=3Y_2+x^4+3x^8+2x^{12}+4Y_2\left(3x^4+3Y_2+3x^8+6x^4Y_2+4Y_2^2\right). \tag{28}$$

Setting $Z_1(t)=x(t)^4+x(t)^3+U(t)$ with $U(t)\in\mathbb{Z}[t]$ and reducing (28) modulo 2 gives the Modulo-8 identity in $\mathbb{F}_2[t]$:

$$\bar{Y}_2(t)=\bar{x}(t)^5+\bar{x}(t)^4+\bar{U}(t)^2+\bar{x}(t)^2\bar{U}(t)=\bar{x}(t)^4\left[\bar{x}(t)+1+\wp\left(\frac{\bar{U}(t)}{\bar{x}(t)^2}\right)\right]\quad \text{in } \mathbb{F}_2[t], \tag{29}$$

where $\wp(f):=f^2+f$ is the Artin–Schreier operator on $\mathbb{F}_2(t)$. Consequently, the formal derivative in $\mathbb{F}_2[t]$ satisfies $\bar{Y}_2'(t)=\bar{x}(t)^2\bar{Z}_1'(t)=\bar{x}(t)^2(\bar{x}(t)^2\bar{x}'(t)+\bar{U}'(t))$, so $\bar{x}(t)^2\mid\bar{Y}_2'(t)$ in $\mathbb{F}_2[t]$.

(ii) ***Artin–Schreier odd-exponent decomposition and Modulo-8 degree jump:*** *For any $\bar{x}(t)\in\mathbb{F}_2[t]$ of degree $m_x=\deg_{\mathbb{F}_2}(\bar{x})\ge 1$, let*

$$\bar{x}(t)+1=\wp(\bar{V}(t))+\bar{x}_{\mathrm{AS}}(t)=(\bar{V}(t)^2+\bar{V}(t))+\sum_{k\ \mathrm{odd}}c_kt^k\quad \text{in } \mathbb{F}_2[t] \tag{30}$$

be the unique Artin–Schreier decomposition obtained by replacing every even-exponent monomial t^{2^rm} (m odd) with t^m, and let $m_{\mathrm{AS}}:=\deg_{\mathbb{F}_2}(\bar{x}_{\mathrm{AS}}(t))$. Whenever $\bar{x}_{\mathrm{AS}}(t)\notin\mathbb{F}_2$ (so $m_{\mathrm{AS}}\ge 1$ is an odd integer), we have for every $Z_1(t)\in\mathbb{Z}[t]$:

$$\deg_{\mathbb{F}_2}(Y_2(t) \bmod 2)\ge 4m_x+m_{\mathrm{AS}}\quad\Longrightarrow\quad \deg Y(t)\ge 4m_x+m_{\mathrm{AS}}+1. \tag{31}$$

In particular, whenever $m_x=\deg_{\mathbb{F}_2}(x(t) \bmod 2)$ is ***odd****, the leading term t^{m_x} cannot be cancelled in $\bar{x}_{\mathrm{AS}}(t)$, so $m_{\mathrm{AS}}=m_x$ holds unconditionally and*

$$\deg_{\mathbb{F}_2}(Y_2(t) \bmod 2)\ge 5m_x\quad\Longrightarrow\quad \deg Y(t)\ge 5m_x+1. \tag{32}$$

(iii) ***Leading-coefficient divisibility in $\mathbb{Z}[t]$ and Modulo-16 congruence in $(\mathbb{Z}/4\mathbb{Z})[t]$:*** *In the Hall–Davenport regime $\deg Y=2\nu,\deg z=3\nu$ with $3\nu>2\deg x$ and $\bar{w}(t)=1$, $\deg_{\mathbb{F}_2}(Z_1 \bmod 2)\le\max(\nu,4m_x)<3\nu$ whenever $3\nu>4m_x$, forcing $4\mid \mathrm{LC}(Y)=a^2$ and $8\mid\mathrm{LC}(z)=\pm a^3$ (i.e., $2\mid a$). Moreover, reducing (28) modulo 4 (Modulo 16 of (1)) gives in $(\mathbb{Z}/4\mathbb{Z})[t]$:*

$$Y_2(t)\equiv-\left(z_1(t)^2+x_0(t)^2z_1(t)\right)+x_0(t)^4+x_0(t)^8+2x_0(t)^{12}+2x_0(t)^2z_2(t)\pmod 4, \tag{33}$$

which depends on $x(t)=x_0(t)+2x_1(t)$ ***solely through*** *$x_0(t)=x(t) \bmod 2$, independent of $x_1(t)$.*

Proof. (i) Expanding $(1 + 2x^4 + 4Y_2)^3 - (x^2 + 2Z_1)^2 - x^4 - 1 = 0$ and dividing by 4 gives (28). Substituting $Z_1 = x^4 + x^3 + U$ into $\bar{Z}_1^2 + \bar{x}^2\bar{Z}_1 + \bar{x}^8 + \bar{x}^4 = \bar{Y}_2$ in $\mathbb{F}_2[t]$ uses $(\bar{x}^4 + \bar{x}^3)^2 + \bar{x}^2(\bar{x}^4 + \bar{x}^3) = (\bar{x}^8 + \bar{x}^6) + (\bar{x}^6 + \bar{x}^5) = \bar{x}^8 + \bar{x}^5$ to give (29). Since $(\bar{f}^2)' = 0$ in $\mathbb{F}_2[t]$, differentiating $\bar{Y}_2 = \bar{Z}_1^2 + \bar{x}^2\bar{Z}_1 + \bar{x}^8 + \bar{x}^4$ gives $\bar{Y}_2' = \bar{x}^2\bar{Z}_1'$.

(ii) Writing $\bar{U}(t) = \bar{x}(t)^2\bar{V}(t) + \bar{W}(t)$ in $\mathbb{F}_2[t]$ transforms (29) into

$$\bar{Y}_2(t) = \bar{x}(t)^4\bar{x}_{\mathrm{AS}}(t) + \bar{W}(t)^2 + \bar{x}(t)^2\bar{W}(t).$$

Here $\deg_{\mathbb{F}_2}(\bar{x}^4\bar{x}_{\mathrm{AS}}) = 4m_x + m_{\mathrm{AS}}$. Since $m_{\mathrm{AS}} \geq 1$ is odd, $4m_x + m_{\mathrm{AS}} > 4m_x$ is an **odd** integer. For any $\bar{W}(t) \in \mathbb{F}_2[t]$ of degree d_W:

- if $d_W > 2m_x$, then $\deg_{\mathbb{F}_2}(\bar{W}^2) = 2d_W > d_W + 2m_x = \deg_{\mathbb{F}_2}(\bar{x}^2\bar{W})$, so $\deg_{\mathbb{F}_2}(\bar{W}^2 + \bar{x}^2\bar{W}) = 2d_W > 4m_x$ is **even** and cannot equal the odd integer $4m_x + m_{\mathrm{AS}}$;
- if $d_W \leq 2m_x$, then $\deg_{\mathbb{F}_2}(\bar{W}^2 + \bar{x}^2\bar{W}) \leq 4m_x < 4m_x + m_{\mathrm{AS}}$ (in fact $\leq 4m_x - 1$ because for $d_W = 2m_x$ the leading terms t^{4m_x} of $\bar{W}^2$ and $\bar{x}^2\bar{W}$ cancel in $\mathbb{F}_2[t]$).

Hence $\deg_{\mathbb{F}_2}(\bar{Y}_2) = \max(4m_x + m_{\mathrm{AS}}, 2d_W) \geq 4m_x + m_{\mathrm{AS}}$. Since $\deg Y(t) = 2\nu$ is even over $\mathbb{R}$ and $\deg Y(t) \geq \deg Y_2(t) \geq \deg_{\mathbb{F}_2}(\bar{Y}_2) \geq 4m_x + m_{\mathrm{AS}}$ (odd), we obtain $\deg Y(t) \geq 4m_x + m_{\mathrm{AS}} + 1$. When m_x is odd, every monomial t^n of $\bar{x}(t)$ with $n < m_x$ reduces to t^m with $m \leq n < m_x$, so t^{m_x} survives in $\bar{x}_{\mathrm{AS}}(t)$ and $m_{\mathrm{AS}} = m_x$.

(iii) Since $(x_0 + 2x_1)^2 = x_0^2 + 4x_0x_1 + 4x_1^2 \equiv x_0^2 \pmod 4$ and $(z_1 + 2z_2)^2 \equiv z_1^2 \pmod 4$, reducing (28) modulo 4 and multiplying by $3 \equiv -1 \pmod 4$ yields (33). □

6.3 Modulo-3 Differential, Modulo-9 Frobenius Carry, and 2-Adic Laurent Obstructions in $\mathbb{Z}[t]$

Theorem 6.4 (Modulo-3 Differential, Modulo-9 Frobenius Carry, and 2-Adic Laurent Obstructions)**.** *Let $(x(t), Y(t), z(t)) \in \mathbb{Z}[t]^3$ be a nonconstant polynomial solution to $Y(t)^3 - z(t)^2 = x(t)^4 + 1$.*

(i) ***Modulo-3 Frobenius and differential identity:*** *Reducing modulo 3 gives $z_0(t)^2 + x_0(t)^4 + 1 = Y_0(t)^3 = Y_0(t^3)$ in $\mathbb{F}_3[t]$ (so every coefficient of $z_0(t)^2 + x_0(t)^4 + 1$ at degree $k \not\equiv 0 \pmod 3$ vanishes in $\mathbb{F}_3[t]$), and differentiating yields $x_0(t)^3x_0'(t) = z_0(t)z_0'(t)$ in $\mathbb{F}_3[t]$.*

(ii) ***Modulo-9 Frobenius carry obstruction (Y_1-elimination modulo 9):*** *Write $x(t) = x_0(t) + 3x_1(t)$, $Y(t) = Y_0(t) + 3Y_1(t)$, and $z(t) = z_0(t) + 3z_1(t)$ with $x_0, Y_0, z_0, x_1, Y_1, z_1 \in \mathbb{Z}[t]$. Since $(Y_0 + 3Y_1)^3 = Y_0^3 + 9Y_0^2Y_1 + 27Y_0Y_1^2 + 27Y_1^3 \equiv Y_0^3 \pmod 9$, $Y_1(t)$ drops out completely modulo 9, and dividing $Y_0(t)^3 - z(t)^2 - x(t)^4 - 1 \equiv 0 \pmod 9$ by 3 yields the exact linear syzygy in $\mathbb{F}_3[t]$:*

$$2z_0(t)z_1(t) + x_0(t)^3x_1(t) \equiv \frac{Y_0(t)^3 - z_0(t)^2 - x_0(t)^4 - 1}{3} \pmod 3 \quad \text{in } \mathbb{F}_3[t]. \tag{34}$$

In particular, at every common factor $\pi(t) \in \mathbb{F}_3[t]$ dividing $\gcd_{\mathbb{F}_3[t]}(x_0(t), z_0(t))$, $\pi(t)$ must divide the carry polynomial $C_3(t) := \frac{1}{3}(Y_0(t)^3 - z_0(t)^2 - x_0(t)^4 - 1) \bmod 3$.

(iii) ***Pointwise modulo-8 and 2-adic Laurent integrality obstructions:*** *For every $t \in \mathbb{Z}$, $Y(t) \equiv 1, 2, 3, 5, 6 \pmod 8$ ($Y(t) \not\equiv 0, 4, 7 \pmod 8$). In the Hall–Davenport regime $d_Y = 2\nu, d_z = 3\nu$ ($3\nu > 2d_x$), $\mathrm{LC}(Y) = a^2$ and $|\mathrm{LC}(z)| = a^3$ for some $a \in \mathbb{Z}_+$ (with $2 \mid a$ whenever $d_x \leq 5$), and the formal Laurent expansion $Y(t)^{3/2} = a^3t^{3\nu}(1 + \sum_{j=1}^{2\nu} \frac{Y_{2\nu-j}}{a^2}t^{-j})^{3/2} = \sum_{m=-\infty}^{3\nu} c_mt^m$ must satisfy $c_m = z_m \in \mathbb{Z}$ for $0 \leq m \leq 3\nu$ and $c_m = 0$ for $4d_x - 3\nu + 1 \leq m < 0$.*

Corollary 6.5 (Immediate Elementary Exclusions in $\mathbb{Z}[t]^3$)**.** *Let $(x(t), Y(t), z(t)) \in \mathbb{Z}[t]^3$ be a nonconstant polynomial solution to $x^4 - Y^3 + z^2 + 1 = 0$.*

(a) ***Odd leading coefficient of*** $Y(t)$***:*** *If* $\mathrm{LC}(Y)$ *is odd, then* $8 \mid \deg Y$ *and* $\lceil 4\deg x/3\rceil \le \deg Y \le 2\deg x - 2$*. In particular,* ***no solution in*** $\mathbb{Z}[t]^3$ ***can have*** $\mathrm{LC}(Y)$ ***odd for any*** $\deg x \le 5$*.*

(b) ***Odd leading coefficient of*** $x(t)$ ***for*** $\deg x = 3$ ***(excluding*** $\deg Y \in \{4,8,10,14\}$ ***by hand):*** *If* $\deg x = 3$ *and* $\mathrm{LC}(x)$ *is odd (in particular, for every monic cubic* $x(t) \in \mathbb{Z}[t]$*), then* $m_x = 3$ *is odd, so Theorem 6.3(ii) forces* $\deg Y \ge 5(3)+1 = 16$ *and* $4 \mid \mathrm{LC}(Y) = a^2$*,* $8 \mid \mathrm{LC}(z) = \pm a^3$*, immediately ruling out* $\deg Y \in \{4,8,10,14\}$ *by pure hand arithmetic and leaving only* $\deg Y \in \{16,20,22\}$*.*

(c) ***Odd leading coefficient of*** $x(t)$ ***for*** $\deg x = 4$***:*** *If* $\deg x = 4$ *and* $\mathrm{LC}(x)$ *is odd* ($x_0(t) = t^4 + a_3t^3 + a_2t^2 + a_1t + a_0 \in \mathbb{F}_2[t]$*), then Theorem 6.2(iv) forces* $\deg Y \ge 16$ *always, and Theorem 6.3(ii) forces* $\deg Y \ge 20$ *whenever* $a_3 \equiv 1 \pmod 2$ *(since* $m_{\mathrm{AS}} = 3$*) and* $\deg Y \ge 18$ *whenever* $a_3 \equiv 0, a_2 + a_1 \equiv 0 \pmod 2$ *(since* $m_{\mathrm{AS}} = 1$*).*

(d) ***Canonical regime*** $(3k,4k,6k)$ ***with odd*** k***:*** *In the canonical regime* $(d_x,d_Y,d_z) = (3k,4k,6k)$*, if* $\mathrm{LC}(x)$ *is odd and* $\mathrm{LC}(z)$ *is even, then* $\mathrm{LC}(Y)^3 = \mathrm{LC}(x)^4 + \mathrm{LC}(z)^2$ *is odd, so* $8 \mid 4k$*, forcing* k *to be* ***even*** *(*$6 \mid \deg x$*).*

Summary of Unconditional $\mathbb{Z}[t]$ Exclusions for $\deg x \le 4$: Combining Theorems 4.2–4.5 with the Modulo-4 Frobenius and Modulo-8 Artin–Schreier Obstructions (Theorems 6.2 and 6.3) completely excludes all polynomial families with $\deg x \le 2$, all families with $\deg x = 3$ and $\deg Y \le 10$ over $\mathbb{Q}[t]$, and additionally excludes $\deg Y = 14$ for $\deg x = 3$ and $\deg Y \in \{6,8,10,12,14\}$ (plus $\deg Y \in \{16,18\}$ when $2 \nmid x_3$) for $\deg x = 4$ in $\mathbb{Z}[t]^3$ whenever $\mathrm{LC}(x)$ is odd.

7 General Conditional Nonexistence Theorems for Arbitrary $\deg x \ge 3$

Finally, we synthesize the geometric and arithmetic obstructions governing arbitrary degrees $d_x = \deg x \ge 3$ into two general conditional theorems that reduce the full polynomial nonexistence conjecture over $\mathbb{Q}[t]$ and $\mathbb{Z}[t]$ to standard principles in Diophantine geometry.

Theorem 7.1 (Conditional Exclusion of the Hall–Davenport Regime for All $d_x \ge 3$)**.** *Assume either of the following standard conjectures for the blown-up rational elliptic surface* $\tilde{S}$ *(Appendix C of [3]):*

(A) ***Strong Polynomial Hall / Function-Field Vojta Inequality with Ramification Control:*** *For any coprime polynomials* $Y(t), z(t) \in \mathbb{Q}[t]$ *of degrees* $2\nu, 3\nu$ *(*$3\nu > 2d_x$*) such that* $Y(t)^3 - z(t)^2 = x(t)^4 + 1$ *factors through* $x(t) \in \mathbb{Q}[t]$ *of degree* $d_x \ge 3$*, the contact order* b_O *with the zero section* O *at* $x = \infty$ *vanishes (*$b_O = 0$*); or*

(B) ***Emptiness of*** $\mathbb{Q}$***-Rational Points on Negative-Codimension Severi Strata:*** *The Severi strata* $\mathcal{V}_{d_x,\nu} \subset |\mathcal{O}_{\tilde{S}}(C)|$ *of rational curves meeting* $O \cap \Theta_0'$ *with contact multiplicity* $b_O = 3\nu - 2d_x > 0$ *along* O *(whose expected dimension is* $3 - \nu < 0$ *for* $d_x = 3, \nu \ge 4$*, and negative relative to the canonical section class for all* $3\nu > 2d_x$*) satisfy* $\mathcal{V}_{d_x,\nu}(\mathbb{Q}) = \emptyset$*.*

Then ***no nonconstant polynomial solution*** $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ ***exists in the Hall–Davenport regime*** $3\deg y = 2\deg z > 4\deg x$ ***for any*** $\deg x \ge 3$*. In particular:*

- *For* $\deg x = 3$*, this excludes the remaining open cases* $\deg y \in \{14,16,20,22\}$ *and completes the proof that* ***no nonconstant polynomial solution exists in*** $\mathbb{Q}[t]^3$ ***with*** $\deg x = 3$ ***for any*** y ***and*** z*.*
- *For arbitrary* $\deg x = d_x \ge 4$ *with* $3 \nmid d_x$*,* ***no nonconstant polynomial solution exists in*** $\mathbb{Q}[t]^3$ ***for any*** y ***and*** z*.*

Theorem 7.2 (Conditional Exclusion of the Canonical Regime $(3k, 4k, 6k)$ for All $k \geq 1$)**.** *For $d_x = 3k$ ($k \geq 1$), every polynomial solution $(x(t), Y(t), z(t)) \in \mathbb{Q}[t]^3$ in the canonical regime $(3k, 4k, 6k)$ corresponds to a $\mathbb{Q}$-rational point on the Severi variety $\mathfrak{B}_k \subset |\mathcal{O}_{X_{12}}(6k)|$ of rational curves of arithmetic genus $p_a(C_k) = 3k^2 + k + 1$ on $X_{12} \subset \mathbb{P}(1,3,4,6)$ having a single unibranch contact of order $3k$ with $\{w = 0\}$ along E_p. Under the arithmetic obstruction $\mathfrak{B}_k(\mathbb{Q}) = \emptyset$ (proved unconditionally for $k = 1$ in Theorem B of [3] via the rank-0 curves $E_0 : z^2 = Y^3 - 1$ and C_R, C_Q), combined with Theorem 7.1,* ***there is no nonconstant polynomial solution*** *$(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$* ***(and hence none in $\mathbb{Z}[t]^3$ or $\mathbb{Z}_+[t]^3$) to $x^4 + y^3 + z^2 + 1 = 0$ for any x, y, z.***

A Summary of Symbolic Verification Pipeline

All symbolic computations marked [C] were executed and verified in exact rational and finite-field arithmetic using SymPy 1.13.3 on Python 3:

1. **Quartic divisorial syzygies for** $\deg x = 4$ **(Theorem 3.1):** Verified by exact symbolic polynomial expansion of $(Y^2 + mY + n)^2 - B^2(Y^3 - x^4 - 1)$ and exact verification of $\Phi_{4,1}(x) \equiv 0$ and $\Phi_{4,2}(x) \equiv 0$.
2. **Covariant syzygy for** $\deg x = 3$ **(Theorem 4.1):** Verified by exact determinant expansion of M_W in $\mathbb{Q}[u, s, \beta, \gamma][T]$ and Hessian/Jacobian determinants of $I(\beta, \gamma) = \beta^3 + s\beta\gamma^2 - u\gamma^3$.
3. **Constant-m and Linear-m branches for** $\nu = 4$ **(Theorems 4.2–4.3):** Exact triangular elimination of $m_0^2, b_0, g_0, g_1^2, W_6, x_0$ in $\mathbb{Q}(\lambda)$, exact verification of $\gcd(P_1(\lambda), P_2(\lambda)) = 1$ and (17), and exact finite-field Newton interpolation and factorization of $\mathrm{Res}_{W_s}(F_{12,1}, F_{23,1})$ (degree 530) over $\mathbb{F}_p[f]$ across 25 primes including $p \in \{10037, 10067, 10133\}$.
4. **Constant-m and Linear-m branches for** $\nu = 5$ **(Theorems 4.4–4.5):** Exact resultant $\mathrm{Res}_\mu(P_6, P_5)$ in $\mathbb{Z}[\lambda]$ for $m_1 = 0$, and exact denominator-free polynomial construction of $\Psi(u) \in \mathbb{Q}[u, B_2, L, \tilde{B}_1, s]$ (3809 terms in 0.51 s) and two-step Euclidean elimination $L = -A_0(\tilde{B}_1, s)/A_1(\tilde{B}_1, s)$ yielding $H_9(\tilde{B}_1, s) = 0$ of degree $(9, 24)$ and G_{I_2}, G_{I_3} of reduced degrees $(47, 95)$ and $(50, 101)$ for $m_1 \neq 0$.
5. **Modulo-8 Artin–Schreier and Modulo-16 identities in** $\mathbb{Z}[t]$ **(Theorem 6.3):** Verified by exact symbolic expansion in $\mathbb{Z}[x, U, Y_2, Y_3, Z_1, Z_2]$ and exhaustive $\mathbb{F}_2[t]$ degree verification across all polynomials $\bar{x}(t) \in \mathbb{F}_2[t]$ of degrees $1 \leq m_x \leq 4$.

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